2016

Q-No.02

(i)

Nominal and ordinal Scale

Nominal scale

Nominal scales are used to classify objects, individuals, groups or even phenomena.

Example

Gender Country

Characteristic:

Nominal scale are mutually exclusive and collectively exhaustive.

Ordinal scale:

Ordinal scale is used when it involves to study, opinion, preference and attitude of people. It is much better then nominal because it provides information to greater extent.

(ii)

Semantic and likert scale

Semantic scale:-

The semantic scale / Differential is a series of attitude scales. This popular attitude - measurement technique consists of presenting and identification of a company's measurement technique consists of presenting an identification of a company, product, brand or other concept, followed by a series of ~~a~~ seven-point bipolar rating scales. Bipolar adjectives, such as "good" and "bad", "modern" and "old - fashioned".

likert scale:

Likert scales measure the degree of any qualitative thing is requires to measure. we can use in likert scale to measure the degree of agreement and Disagreement on a particular statement.

likerT scale has a minimum of Two categories Such as agree and Disagree but it is usually better To use four To eight categories.

The categories can be increase at the end by adding.

"strongly agree"

"Some what agree"

"agree"

(iii)

Cross-sectional and longitudinal study

Cross-sectional study:-

Cross sectional research:- 2016-(c)

→ It is a type of research that takes place at a single point in time

→ In such study the data are gathered just once.

→ In this study several groups of people are examined at one point in time.

→ It is like taking a slice or cross section that we are measuring or observing.

longitudinal research:- 2016 (c)

→ It is a type of study that takes place over time.

→ In such studies researcher want to study people or phenomena at more than on point in time.

→ when the Data on dependent variable are gathered at two or more points in time then it is called longitudinal research

→ It acquires more time and cost than cross sectional study.

Closed-Ended Questions:- 2016 (d)

→ In a closed question the possible answers are set out in the Questionnaire

→ closed-ended questions have short check mark responses.

→ There is no provision other than the given answers for the respondent.

→ The respondent or the investigator ticks the category that best describes the respondents answer.

→ closed- Questions can also be used in interviews as well as Questionnaire.

→ These are widely used Types of Questions

→ These may also known as multiple choice Questions.

For Example:-

→ what is Your Designation?

☐ lecture ☐ Assistant prof ☐ Associate prof

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Simple Random Sampling:

In simple random sampling, there is an equal probability of each population and each sample of being selected. In simple random sampling researcher is always interested to increase efficiency in sampling. This sampling technique has a quality to increase efficiency. Researcher uses the following technique to increase efficiency.

1. Holding accuracy and decreasing cost.
2. Holding cost constant and increasing accuracy.
3. Rate of increasing the accuracy should be faster than the rate of cost.
4. Rate of decreasing the accuracy should be at a slower rate than the rate of cost.

Note: Sampling efficiency is the ratio of accuracy over cost.

It is very convenient process of sampling. It has an equal chance of probability of selecting the elements of population e.g.

If we want analyze the selection of teachers, what are necessary grounds.

We want to study this project from survey of different colleges. Then there is chance of selecting all the students from same college or there is chance that every college has it representation equally.

Some experts opinion is to chose 10 percents of population when sample size is small.

According to Ralph L. Rosnow:

> *Random Sampling is a sample chosen by ghane procedure and with known probabilities of selection so that every individual in the population will have the same likelihood of being selected.*

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Stratified Sampling:

Stratified sampling has a quality to increase the sampling efficiency by increasing the accuracy at faster rate than the cost. Both accuracy and cost increase. Due to variables used to form groups stratification is almost efficient statistically than simple random sampling and at worst is equal to it. It is used where target is know the features of certain subgroups of population it is also useful when different methods of observation are under consideration.

Systematic Sampling: 2016

It has equal chance of selecting of all the population in sample. But it requires proper formula or technique or planning. In this type researcher remains impartial but for researcher it is necessary to have complete list of all the elements of population. e.g.,

From the strength of 50 students in a class if we want to select 5 students then the students are divided into 5 groups. It means list is distributed of every 10 students and then making the list of first 10 students then next students and so on. Finally from these five lists of consist of 10 students, draw is made of one student from each list. Then there is equal chance of students to be selecting for sample from five lists.

Q.3 Explain the errors that effect accuracy. also give the reasons how they can be avoided

There are several types of errors that can affect the accuracy of a system or model, including:

1. **Sampling errors:** This occurs when the data used to train the model is not representative of the entire population. To avoid this, ensure that the sample size is large enough and that the data is collected from a diverse range of sources.
2. **Measurement errors:** This occurs when there are errors in the measurement or recording of data. To avoid this, ensure that the measurement instruments are calibrated and that the data is recorded accurately.
3. **Human errors:** This occurs when there are errors in the data entry or processing by humans. To avoid this, use automated data entry processes and ensure that the data is reviewed and validated by multiple people.

4. **Modeling errors:** This occurs when there are errors in the assumptions or methods used to create the model. To avoid this, ensure that the modeling techniques used are appropriate for the data and that the assumptions made are valid.
5. **Data preprocessing errors:** This occurs when there are errors in the preprocessing of the data, such as missing values or incorrect normalization. To avoid this, ensure that the data is preprocessed carefully and that any missing values are handled appropriately.
6. **Overfitting errors:** This occurs when the model is too complex and fits the training data too closely, resulting in poor performance on new data. To avoid this, use techniques such as regularization to prevent overfitting and ensure that the model is evaluated on independent test data.
7. **Underfitting errors:** This occurs when the model is too simple and does not capture the complexity of the data, resulting in poor performance on both the training and test data. To avoid this, use more complex models or consider using different features or preprocessing techniques.

Overall, to avoid errors that affect accuracy, it is important to carefully design the study or model, use appropriate techniques, and validate the data and results through independent evaluation.